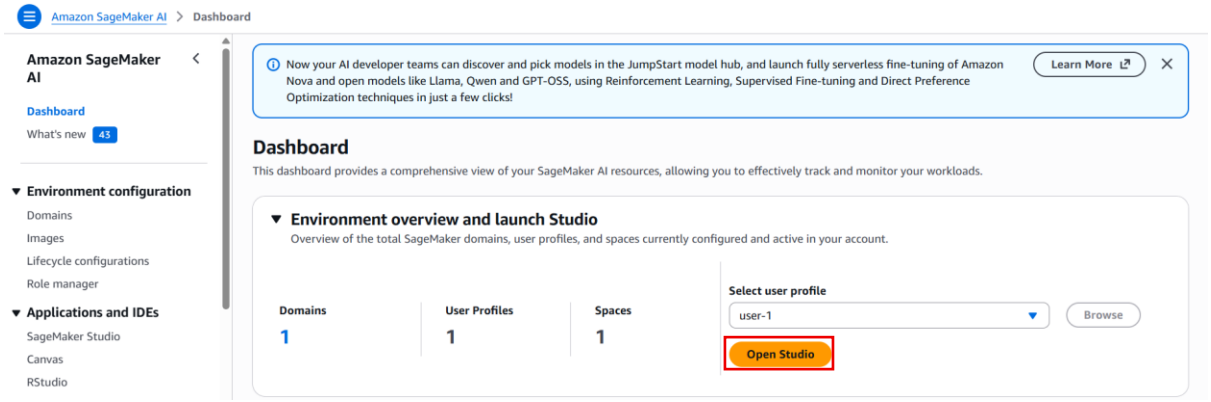
Import Data Using Data Wrangler

To Begin with the Lab

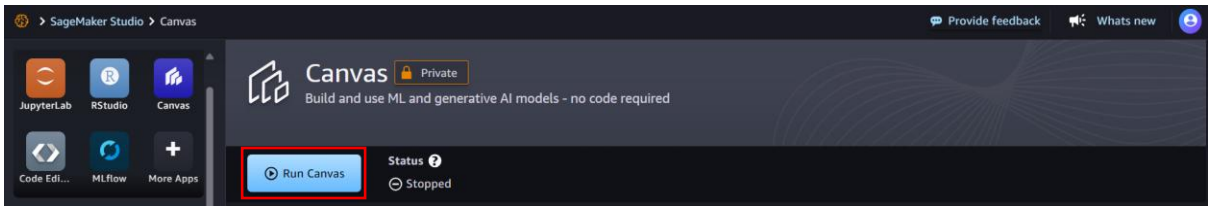
Summary of the Lab

This lab demonstrates how to import and prepare a CSV dataset using Amazon SageMaker Data Wrangler within Canvas. After logging into AWS and launching SageMaker Studio, you access Canvas, start Data Wrangler, and create a new data flow. The dataset is uploaded locally, previewed for structure and quality, and optionally sampled for efficiency. After verifying file settings and data consistency, the dataset is imported, creating a visual data flow with a source node and auto-detected data types, ready for further transformations.

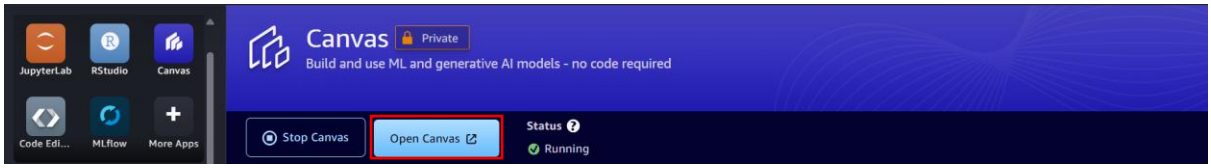
- Prerequisites
  - SageMaker Domain already configured
  - User profile created
  - Access to SageMaker Studio / Canvas
  - CSV dataset available on local machine
- Log in to AWS Console.
- Open Amazon SageMaker.
- Launch your user profile.



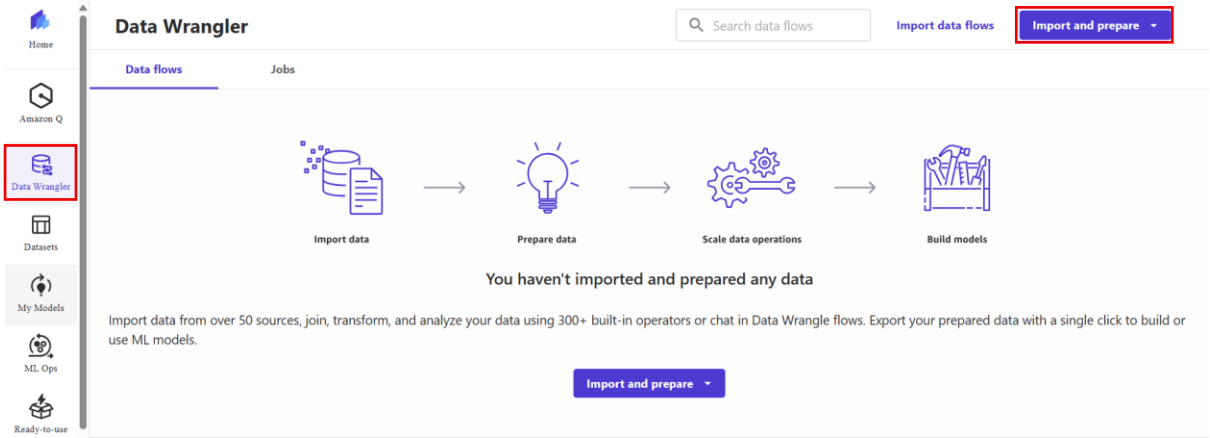
- From Studio applications, locate Canvas.
- If status is Stopped, click Start.



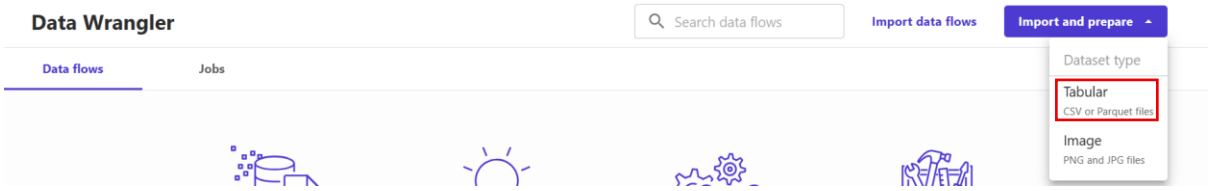
- Wait until status changes to Running.
- Click Open Canvas.



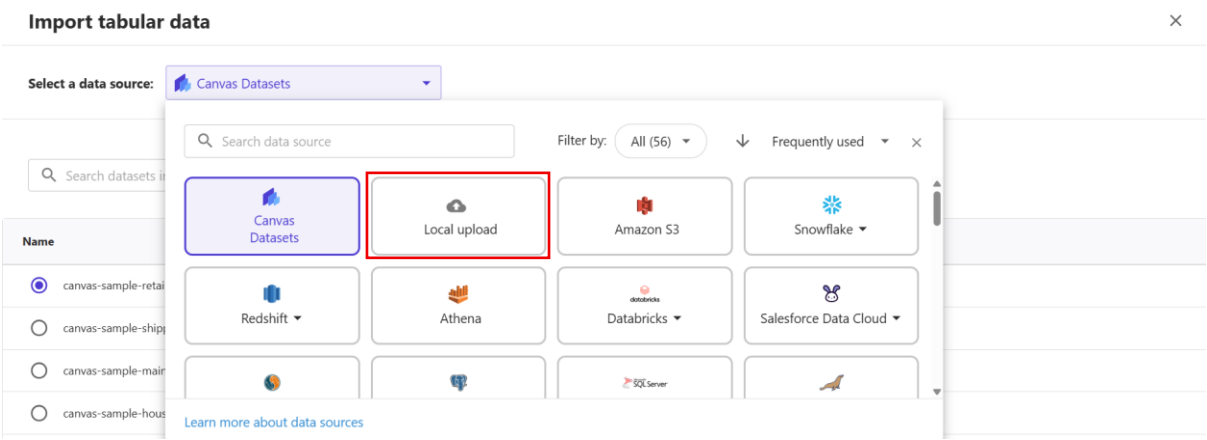
- Inside Canvas, use the left navigation panel.
- Select Data Wrangler.
- In Data Wrangler, choose Import and prepare.
- Start a new data flow.



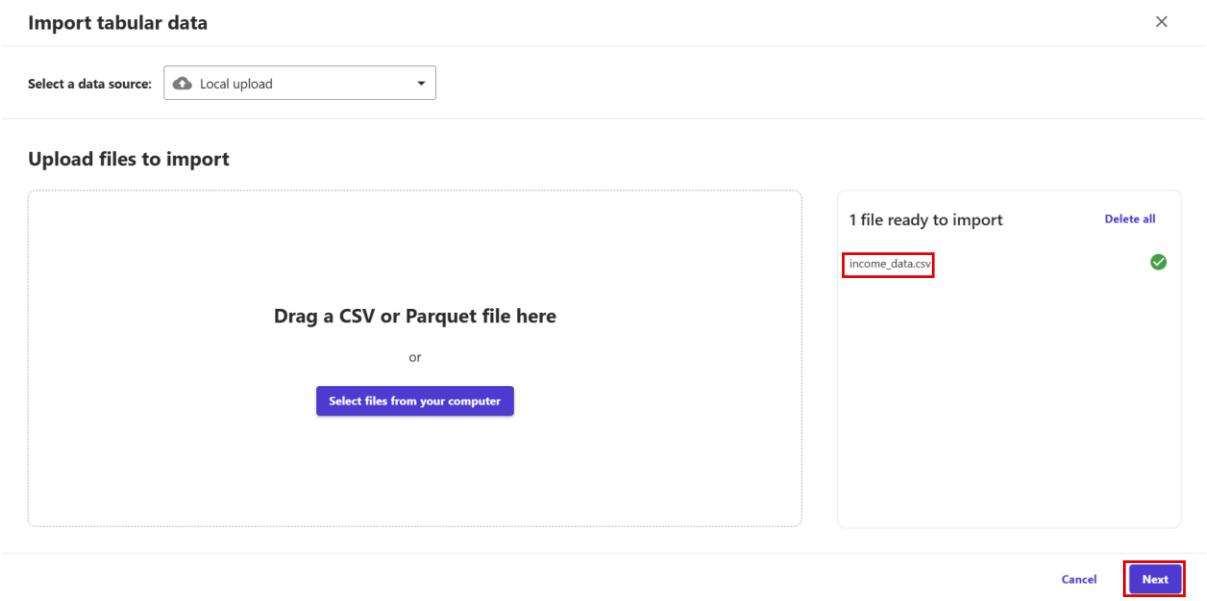
- Choose Tabular, as we have data in table(CSV format).



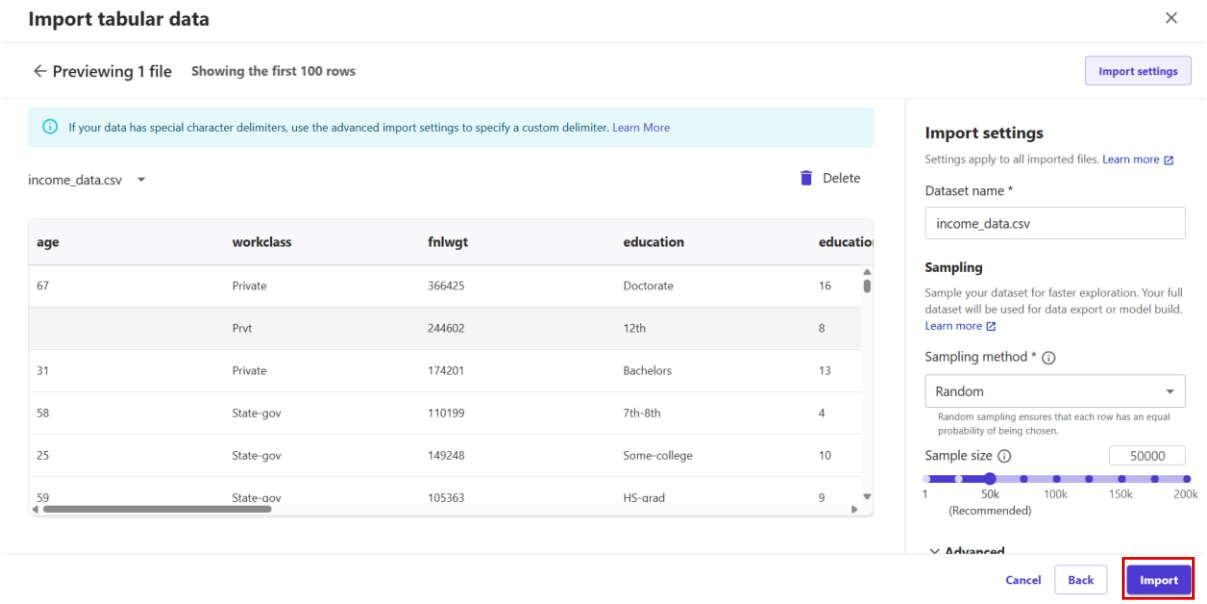
- Select Local Upload.



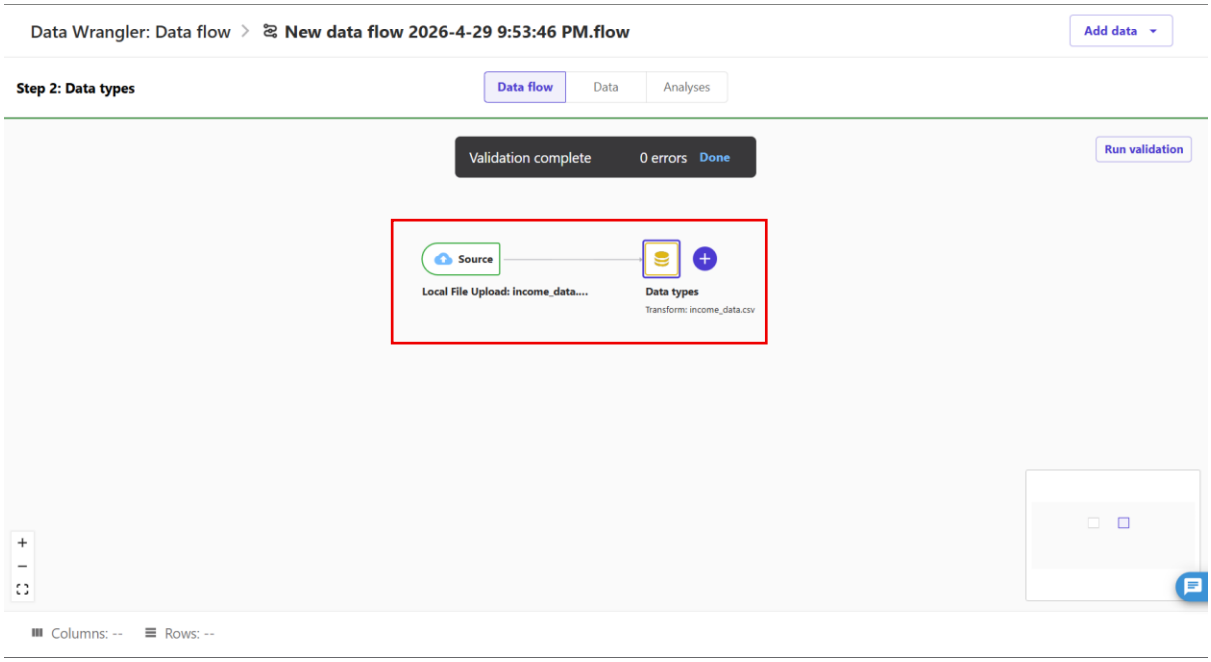
- Drag and drop your CSV file into the upload window.
- Verify file appears successfully.
- Click Next.



- Review the dataset preview to confirm columns, data types, missing values, and spot issues like invalid ages, text in numeric fields, null cells, or inconsistent category names.
- If the dataset is large, enable Random Sampling (~50,000 rows) for faster processing, then verify detected file options such as encoding, delimiter, header row, and column separation.
- Click Import.



- After import, confirm visual data flow shows:
- Source Node (CSV dataset)
- Auto-detected data type step
- Ready for transformation steps


Data Wrangler: Data flow
New data flow 2026-4-29 9:53:46 PM.flow
Add data
Step 2: Data types
Data flow
Data
Analyses
Validation complete
0 errors
Done
Run validation
Source
Local File Upload: income_data....
Data types
Transform: income_data.csv
Columns: --
Rows: --